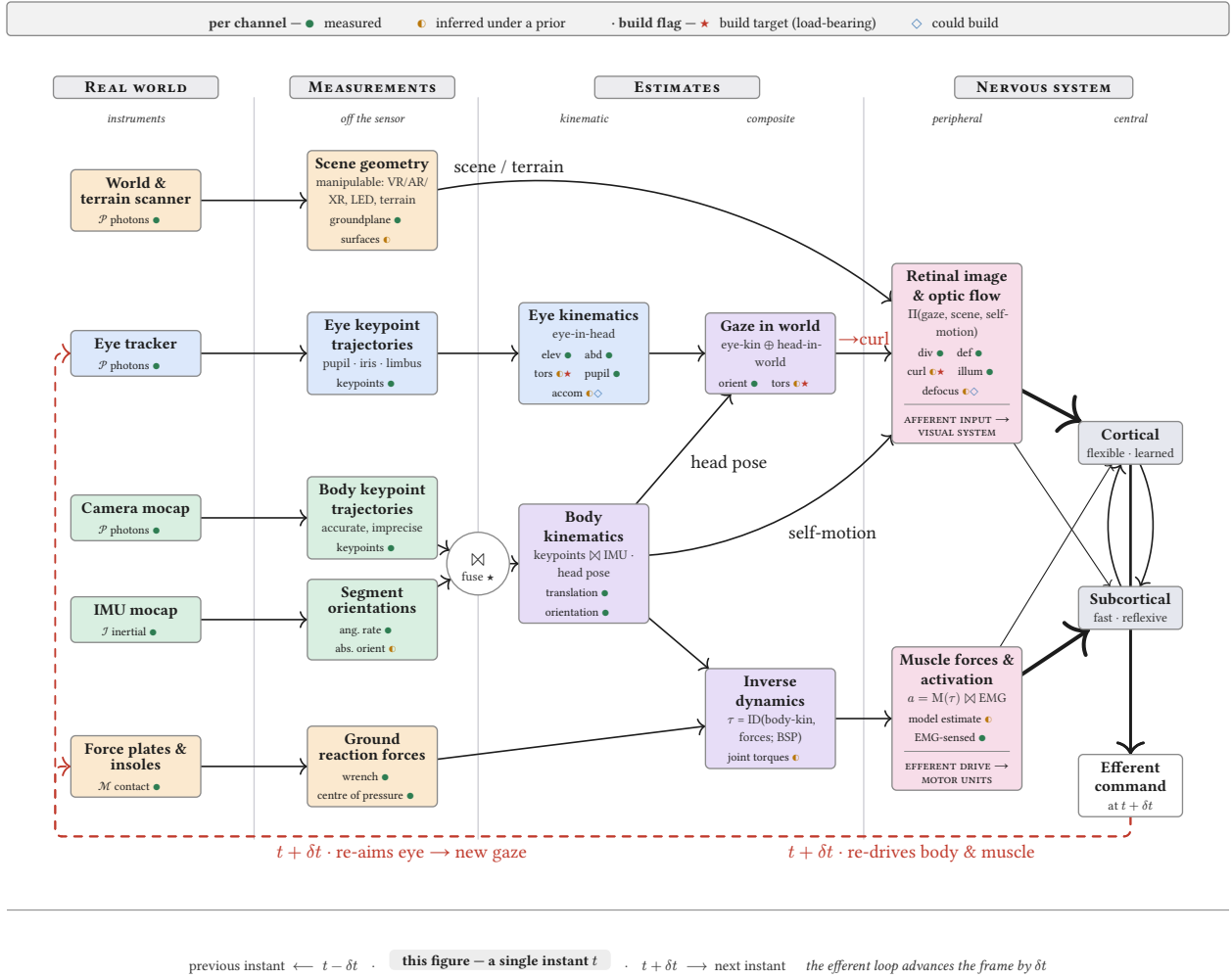




READING THE CHART AS A CALCULUS		
each value is a bundle of channels (degrees of freedom); the top legend grades each channel.		
Notation	The chains — energy → sensor → measurement → estimate	Into the nervous system
$\mathcal{E}$ an energy a sensor is sensitive to: $\mathcal{P}$ photons · $\mathcal{I}$ inertial · $\mathcal{M}$ contact	Eye $\mathcal{P} \vdash$ eye keypoints → eye-kin (elev, abd, tors, pupil, accom)	Retinal input $\rho = \Pi(g, \text{scene}, \dot{x}) \approx \rho^*$ retinal image & optic flow (afferent)
$S : \mathcal{E} \vdash v$ transduce — sensor S turns energy $\mathcal{E}$ into a value v	Body $\mathcal{P} \vdash$ body keypoints; $\mathcal{I} \vdash$ segment orient; body-kin = keypoints ⊔ orient (head pose falls out)	Kinetics $\mathcal{M} \vdash$ GRF; $\tau = \text{ID}(\text{body-kin}, \text{GRF}; \text{BSP})$
$\vdash$ exposes / yields (the box on the left produces the value on the right)	Gaze $g = \text{eye-kin} \oplus \text{head-in-world}$	Motor $a = M(\tau) \boxtimes \text{EMG} \approx a^*$ muscle drive (efferent)
$\rightarrow$ feeds — becomes an input to the next step		
$\oplus$ compose frames — chain relative poses; carries channels through unchanged		
$\boxtimes$ fuse — union the measured subspaces of two estimators		
Π project — the only operator that maps input DOF onto output channels		
$\approx$ approximates the true latent value		

Figure 1: **The measurement chain of a DOME.** Reading across four bands — **real world** → **measurements** → **estimates** → **nervous system**. Sensors transduce physical energies into measurements (one step off the sensor); fusion ($\boxtimes$) and frame composition ($\oplus$) build the kinematic and composite estimates that project (Π) onto the two points where the body meets the nervous system — the retinal stimulus entering vision and the muscle drive leaving the motor system. Every box is a bundle of **channels**; each channel is marked on two axes — measured (●) or inferred under a prior (●) now, and, if unmeasured, a committed build target (★) or merely feasible (◇). The eye path makes the argument concrete: ocular torsion is inferred under Listing's law today and corrupts exactly the retinal **curl** channel — which is why it is a build target. The two peripheral signals project to the **central nervous system** (cortical + subcortical, reciprocally coupled), whose efferent command re-aims the eye and re-drives the body at the next instant — closing the loop. The whole figure is a **single instant t** : it is preceded by the infinitesimal before it and followed by the one after, and the dashed red loop is what advances the frame by δt . The calculus strip beneath the diagram gives the full reading rule, every symbol defined.